Utrecht University

Predicting Complications Following Cancer Surgery

A preoperative prediction model for postoperative complications following oesophageal cancer surgery

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Abstract

Surgery for oesophageal cancer is curative but carries a high risk of postoperative complications that shorten survival, so reliable preoperative risk prediction could help target prehabilitation. Using the Dutch Upper Gastrointestinal Cancer Audit for 1,022 oesophageal cancer patients treated at Amsterdam University Medical Center between 2016 and 2025, this study asks how far complications can be predicted from routinely available information, and whether the operative course or the preoperative physiotherapy assessment raises that ceiling. Penalised logistic models were evaluated by cross validation for discrimination, calibration, and clinical utility, following the TRIPOD+AI reporting guideline, and compared against an ASA based baseline, a lasso, and a random forest. The preoperative record predicted the principal outcomes only modestly, with areas under the curve between 0.55 and 0.60. The operative course raised discrimination for major complications, from 0.594 to 0.638, but for no other outcome. The physiotherapy risk conclusion and the functioning coded from clinical notes tracked risk and agreed with one another, yet added nothing beyond the surgical model, and flexible machine learning did not improve on the logistic baseline. Calibration was adequate for several outcomes but poor for pulmonary complications, where the model was overconfident. The contribution is a transparent and well validated account of how far routine information can be taken for this problem, showing that accurate short-term prediction will require richer data rather than a more complex model.

Glossary

The clinical terms and abbreviations below appear throughout the thesis and are explained in plain language for readers from outside the medical field.

Clinical Term	Meaning
Oesophagectomy	The operation to remove part or all of the oesophagus, the main treatment for oesophageal cancer.
Neoadjuvant therapy	Chemotherapy or chemoradiotherapy given before surgery to shrink the tumour.
Clinical T stage and N stage	The size and local extent of the tumour (T) and whether the cancer has reached nearby lymph nodes (N), the two components of cancer staging recorded before surgery and used here as predictors.
Adenocarcinoma	One of the two main tissue types of oesophageal cancer, the other being squamous cell carcinoma; the tumour type is recorded as a preoperative characteristic.
Anastomosis	The surgical join made after the diseased section is removed; a cervical anastomosis is made in the neck.
Conversion to open	Switching during the operation from keyhole (minimally invasive) surgery to a larger open incision.
Minimally invasive resection	Keyhole surgery through small incisions with a camera, as opposed to open surgery through one large incision.
Transthoracic resection	An approach that reaches the oesophagus by opening the chest, as opposed to a route through the abdomen and diaphragm.
Salvage resection	Surgery performed after earlier definitive chemoradiotherapy has failed; excluded from the cohort.
Prophylactic resection	Surgery performed to prevent disease rather than to treat an existing tumour; excluded from the cohort.
Clavien-Dindo classification	A graded scale of how serious a surgical complication is; grade IIIb or higher is treated here as a major complication.
Anastomotic leakage	Leakage at the surgical join where the oesophagus was reconnected, one of the most serious complications after this surgery.
Reintervention	A further procedure or operation needed to treat a complication after the original surgery.
Charlson Comorbidity Index	A weighted count of chronic illnesses that summarises how much other disease a patient carries.
Textbook outcome	A composite of an ideal recovery: no severe complication, acceptable length of stay, and no readmission or early death.
Prehabilitation	Structured exercise and lifestyle preparation before surgery to improve fitness and recovery.

Abbreviation	Meaning
A-PROOF	The research project that developed the functioning classifiers used here, which read patient functioning from clinical notes.
ASA	American Society of Anesthesiologists physical status, a five-point grade of a patient's overall fitness for surgery.
COPD	Chronic obstructive pulmonary disease, a long-term lung condition recorded in the registry as chronic pulmonary disease and examined here as a risk factor.
DUCA	Dutch Upper Gastrointestinal Cancer Audit, the national registry that provided the cohort.
ECCG	Esophagectomy Complications Consensus Group, which set the standard definitions for complications after oesophageal surgery.
ICF	International Classification of Functioning, Disability and Health, the World Health Organization framework for describing functioning.
METc	Medical Ethics Committee (Medisch Ethische Toetsingscommissie), the body that reviewed the study and issued the non-WMO waiver under which it was conducted.
POSSUM	Physiological and Operative Severity Score for the enumeration of Mortality and morbidity, an established surgical risk score.
TRIPOD+AI	A reporting guideline for prediction models built with regression or machine learning, followed throughout this thesis.
WMO	Dutch Medical Research Involving Human Subjects Act; this study qualified for a non-WMO waiver.

1. Introduction

1.1. Clinical context and motivation

Surgery for oesophageal cancer offers the prospect of cure but carries a high risk of postoperative complications. Pulmonary complications, anastomotic leakage, cardiac events, and infections are common, and they are more than perioperative inconveniences. An individual patient data meta-analysis of 2,181 patients reported that severe postoperative complications after oesophagectomy were associated with an absolute reduction of approximately 8.6 months in mean survival at sixty months (Bona et al., 2024). Pulmonary complications in particular carry the largest oncological effect, with a pooled hazard ratio of 1.37 for overall survival, and they are among the most frequent severe complications (Booka et al., 2018). Reliable preoperative prediction of these complications has nevertheless proven difficult to establish (Avendano et al., 2002). Reducing complications is therefore a clinical objective with direct consequences for survival, not only a matter of care quality.

Because the burden of complications is concentrated around the time of surgery, the preoperative period has become the focus of efforts to identify and reduce risk. At Amsterdam UMC, the Department of Rehabilitation Medicine has operated a preoperative screening and prehabilitation pathway since 2007. Physiotherapists assess patients before surgery and triage those with reduced fitness toward a structured prehabilitation programme built around modifiable factors such as physical activity, nutrition, body weight, and the cessation of tobacco and alcohol use. The premise of the pathway is that patients identified as high risk before surgery can be prepared so that their likelihood of complications is reduced. The value of the pathway therefore depends on how accurately preoperative risk can be assessed, and on whether information collected during routine care can support that assessment.

1.2. Background and knowledge gap

Preoperative risk assessment for major surgery is an established field. General purpose instruments such as the American Society of Anesthesiologists physical status classification and the Charlson Comorbidity Index summarise a patient's fitness and comorbidity burden into a single score, and surgical risk models combine physiological and operative variables to estimate the probability of morbidity and mortality. These instruments are widely used because they are simple and rely on information that is routinely available.

Their performance in oesophageal cancer surgery is nevertheless limited. A nationwide study using the Dutch Upper Gastrointestinal Cancer Audit found that age and the Charlson Comorbidity Index were not independent predictors of severe complications after curative oesophagectomy and identified the ASA score and the surgical approach as the relevant risk factors instead (Baranov et al., 2022). This finding is important because the same registry, which records detailed preoperative and operative information for patients undergoing oesophageal cancer surgery in the Netherlands, is exactly the kind of large and routinely collected data source that modern prediction methods are designed to exploit. The scientific gap is therefore specific. Routinely collected information is abundant, yet whether it can be combined into a model that predicts complications well enough to support preoperative decision making in this population remains unresolved.

The societal dimension of this gap is concrete. Patients who would benefit from intensified prehabilitation may not be identified by current screening, while others may undergo preparation they do not need. A prediction model built on information already gathered during routine care, requiring no additional tests and imposing no extra burden on patients, could help to target prehabilitation more precisely. This is relevant both to patient outcomes and to the efficient use of limited perioperative care capacity, since the complications being targeted are the same ones shown above to shorten survival.

Three distinct layers of routinely collected information are available for this population, and they form the structure of the present study. The first is the structured preoperative record in the surgical registry. The second is the operative course, the surgical approach, the anastomosis, and events such as conversion from a minimally invasive to an open procedure, which is recorded for every patient and may carry information that the preoperative record does not. The third is the preoperative physiotherapy assessment, which since 2007 has produced a risk conclusion for each screened patient, and which can be enriched by a set of classifiers, developed within the A-PROOF project together with the Computational Lexicology and Terminology Lab at Vrije Universiteit Amsterdam, that extract functioning coded according to the International Classification of Functioning, Disability and Health from free text clinical notes (Meskers et al., 2022).

The question that organises this thesis is how far complications can be predicted from the preoperative record, and whether the operative course or the physiotherapy assessment raises that ceiling. No prior study appears to have tested these three layers of routine information together, under a calibration first reporting standard with decision curve analysis, for these outcomes in this population. The contribution is therefore the first calibrated, decision-curve-evaluated account of whether the operative course or the preoperative physiotherapy assessment moves the preoperative prediction ceiling for these outcomes, establishing not only how far routine data can be taken but which of these readily available layers is worth collecting for prediction at all.

1.3. Research questions

Translated into a data science problem, the clinical objective becomes a set of binary classification tasks. Using information available at a defined point in the care pathway, the aim is to estimate the probability that a patient will experience a particular postoperative outcome. The performance of such a model must be judged on two axes together, discrimination, the ability to rank patients by risk, and calibration, the agreement between predicted and observed risk, because a model intended to inform clinical decisions must be both accurate and trustworthy. Preference is given throughout to models whose predictions can be explained in clinical terms. This study focuses on the oesophageal cohort within the registry; the reasons are set out in Section 3.2. The central question of this study is the following.

How far can complications after oesophageal cancer surgery be predicted from routinely available information, and does adding the operative course or the preoperative physiotherapy assessment raise that ceiling?

This central question is addressed through four sub questions, each examining one layer of the available information and each answerable with the data at hand.

- *SQ1. How well can preoperative factors predict the principal postoperative outcomes, namely major complications defined as Clavien-Dindo grade IIIb or higher, prolonged length of hospital stay, and pulmonary complications?*
- *SQ2. Which factors carry the predictive signal, across both preoperative and operative information, and are they consistent with established clinical risk factors?*
- *SQ3. Does information from the operative course add predictive value beyond what is available before surgery?*
- *SQ4. Does the preoperative physiotherapy assessment, both the risk conclusion and the ICF coded functioning extracted from clinical notes, add predictive value beyond a surgical model?*

The four sub questions form a single line of inquiry. SQ1 establishes the predictive ceiling of the structured preoperative record and reports it honestly through discrimination and calibration. SQ2 examines the drivers of the prediction, which speaks both to clinical plausibility and to the interpretability that clinical adoption requires. SQ3 and SQ4 then test whether two further layers of routine information, the operative course and the physiotherapy assessment, move that ceiling. Throughout, a transparent baseline using the ASA score and an interpretable logistic model are compared against more flexible methods, so that the additional complexity of machine learning is only adopted if it earns its place.

1.4. Ethical considerations

Because this study uses sensitive patient data, several ethical considerations frame the work. All data were processed within a secure research environment, and patient level data were not exported from it. The study received a non-WMO waiver from the Medical Ethics Committee (METc number 2020.277), indicating that it did not fall under the Dutch Medical Research Involving Human Subjects Act, and the analysis was conducted under the data protection regime governing the registry, with patients described only in aggregate so that no individual can be identified.

Two further considerations concern the responsible use of a prediction model. A model that discriminates poorly but is presented as a risk tool could give false reassurance and lead to undertreatment of patients who are in fact at risk. For this reason, the analysis emphasises honest reporting of uncertainty and calibration rather than headline discrimination alone. In addition, predictive performance can differ between subgroups defined by age, sex, or tumour stage. Because this study deliberately reports overall performance and does not develop a tool intended for clinical use, a formal subgroup fairness assessment is a prerequisite for any future deployable model rather than a step taken here, and it is noted as such.

1.5. Outline of the thesis

The remainder of this thesis is organised as follows. Chapter 2 reviews the clinical and data science literature on complications after oesophageal cancer surgery, on the operative course, on prediction modelling in this setting, and on the extraction of functioning from clinical notes, and situates the present work within it. Chapter 3 describes the data sources, the cohort selection, the preparation of the predictors and the outcomes, the physiotherapy assessment features, and the modelling and validation strategy. Chapter 4 presents the results, organised around the four sub questions. Chapter 5 discusses the findings, their limitations, and their clinical and ethical implications, identifies directions for future work, and concludes.

2. Literature Review

2.1. Complications after oesophageal cancer surgery

Surgery remains central to the curative treatment of oesophageal and gastric cancer, yet it carries substantial morbidity. A systematic review and meta-analysis of patient related prognostic factors reported that major complications occur in approximately 25 per cent of patients after oesophagectomy, with short term mortality of around 4 per cent (van Kooten et al., 2022). Severe complications, defined as Clavien-Dindo grade above three, were observed in close to 30 per cent of patients in an individual patient data meta-analysis (Bona et al., 2024). To allow comparison across studies and centres, the severity of complications is commonly graded with the Clavien-Dindo classification (Dindo et al., 2004), and the Esophagectomy Complications Consensus Group standardised the definitions of individual complications such as pneumonia and anastomotic leakage (Low et al., 2015). Building on these definitions, the textbook outcome combines several desirable short-term endpoints, including the absence of severe complications, an acceptable length of stay, and the absence of readmission or early mortality, into a single composite quality measure for oesophagogastric surgery (Busweiler et al., 2017).

The clinical importance of these outcomes extends well beyond the surgical admission. Pooled evidence shows that postoperative complications shorten long term survival, with pulmonary complications associated with a hazard ratio of 1.37 for five-year overall survival and anastomotic leakage with a hazard ratio of 1.20 (Booka et al., 2018). The individual patient data meta-analysis quantified this loss in absolute terms, with patients experiencing severe complications surviving approximately 8.6 months less over five years of follow up (Bona et al., 2024). Achieving a textbook outcome, by contrast, is associated with better overall and disease-free survival (Kalff et al., 2021). In the Netherlands these outcomes are recorded nationally through the Dutch Upper Gastrointestinal Cancer Audit, which has tracked complications, length of stay, readmission, and mortality across eight years of surgical auditing (Voeten et al., 2021).

Taken together, this body of work establishes that complications after oesophageal cancer surgery are frequent, can be graded in a standardised way, and carry consequences for survival rather than only for the index admission. This makes the preoperative identification of patients at risk a clinically meaningful objective, and it motivates the outcomes examined in this study, major complications at Clavien-Dindo grade IIIb or higher, prolonged length of stay, and pulmonary complications, with reintervention and the textbook outcome as further endpoints.

2.2. Pulmonary complications and their predictors

Among the complications that follow oesophagectomy, pulmonary complications are among the most frequent and carry the largest effect on survival (Booka et al., 2018; Manara et al., 2024). The search for reliable preoperative predictors has a long history. An early and frequently cited study concluded that, although impaired lung function raised the risk, there were no good preoperative predictors of pulmonary complications, with patients whose forced expiratory volume in one second fell below 65 per cent of predicted appearing to be at greatest risk (Avendano et al., 2002). That conclusion has proven durable, and it frames the question that this study revisits with a larger cohort and modern methods.

More recent evidence from the Dutch registry complicates the assumption that comorbidity burden is the main driver of risk. In a population-based cohort drawn from the national audit, age and the Charlson Comorbidity Index were not independent risk factors for severe complications after curative oesophagectomy, whereas the ASA score and the surgical approach were (Baranov et al., 2022). This finding matters for the present study in two ways. It suggests that a single fitness summary such as the ASA score may carry much of the available preoperative signal, and it cautions against assuming that a detailed comorbidity index will necessarily improve prediction. Both points are examined directly in the second sub question.

2.3. Existing preoperative risk scores and their limitations

Several preoperative risk instruments are used in surgical practice. The ASA physical status classification summarises overall fitness on a simple ordinal scale, the Charlson Comorbidity Index aggregates weighted comorbidities into a single number, and the O-POSSUM score combines physiological and operative variables specifically for oesophagogastric surgery (Tekkis et al., 2004). The performance of these instruments outside their development setting does not hold up well. The original O-POSSUM development study reported an area under the curve of approximately 0.80 for operative mortality (Tekkis et al., 2004), yet independent validation found that it overpredicted mortality and discriminated poorly between patients who survived and those who did not (Nagabhushan et al., 2007). Two limitations recur across this literature. First, discrimination that appears strong in a development sample degrades on independent data, and it is weaker for morbidity outcomes such as anastomotic leakage than for mortality. Second, calibration, the agreement between predicted and observed risk, is frequently poor and is often not reported at all. These two limitations define the scientific gap that the present study addresses.

2.4. Prediction modelling in surgical oncology

A growing body of work applies prediction modelling to postoperative outcomes in surgical oncology. Most directly relevant, a Dutch study built machine learning models to predict anastomotic leakage and pneumonia after oesophagectomy from preoperative and early postoperative data in a cohort of 417 patients, demonstrating the feasibility of registry style prediction in this exact clinical setting (van de Beld et al., 2024). That study, however, modelled two specific complications in a single centre cohort of modest size, did not incorporate physiotherapy or functional information, and did not place calibration or clinical utility at the centre of its evaluation. It therefore leaves open whether a broader set of outcomes can be predicted from routine information in a larger cohort, whether the preoperative physiotherapy assessment adds anything, and how well such models are calibrated, which are the questions taken up here.

In gastric cancer surgery, machine learning has been applied to predict surgical outcomes (Lu et al., 2019), and prediction of prolonged length of stay has been studied in lung resection by developing and comparing several machine learning models (Zhang et al., 2024). Beyond individual models, the systematic review by van Kooten et al. (2022) synthesised a large set of prognostic factors for anastomotic leakage, major complications, and mortality, and identified cardiac comorbidity, male sex, diabetes, advanced age, alcohol use, and body mass index as recurring patient related predictors. This established set provides the clinical benchmark against which the factors identified in the present study can be compared in the second sub question. The recurrence of the same predictors across many studies also suggests that the preoperative signal for these outcomes, while real, is limited and already well characterised, which tempers expectations about how much any single new model can improve upon it.

2.5. The operative course as a determinant of outcome

The registry records not only the preoperative state but also the operative course, and the literature gives reason to expect that this layer carries genuine prognostic information. The same Dutch registry study that dismissed age and comorbidity found the surgical approach to be an independent risk factor for severe complications (Baranov et al., 2022), and the standardised complication definitions for this surgery place anastomotic leakage, which is closely tied to the location and construction of the anastomosis, among the most consequential events (Low et al., 2015). The route of resection, whether the operation is completed minimally invasively or converted to an open procedure, and the location of the anastomosis are therefore plausible determinants of outcome that are not known before surgery. Whether they add predictive value beyond the preoperative record, however, has not been established for these outcomes in this population, which is the question posed in the third sub question.

2.6. Machine learning versus transparent baselines

A central question for any applied prediction study is whether flexible machine learning methods outperform transparent regression models. The most influential evidence is a systematic review of 282 model comparisons, which found that among studies at low risk of bias the difference in performance between logistic regression and machine learning, expressed on the logit area under the curve scale, was 0.00, with a 95 per cent confidence interval from minus 0.18 to 0.18 (Christodoulou et al., 2019). The same review found that apparent advantages for machine learning were concentrated in studies with methodological weaknesses, that 68 per cent of studies showed signs of validation bias, and that calibration was not addressed in 79 per cent of them. Against this background there is no firm basis to expect that flexible models will substantially outperform a well specified logistic regression in the present setting, and a finding that they do not would be consistent with the wider literature rather than disappointing.

How such comparisons should be conducted is now codified. The TRIPOD plus AI reporting guideline provides a common standard for clinical prediction models built with either regression or machine learning, and emphasises transparent reporting of model development, validation, and calibration (Collins et al., 2024). Two further methodological findings shape the analytical strategy of this study. First, correcting class imbalance through resampling has been shown to harm calibration by systematically overestimating risk, without improving discrimination, which argues against such corrections for the imbalanced complication outcomes studied here (van den Goorbergh et al., 2022). Second, interpretable models can match the accuracy of black-box (opaque) models while exposing clinically meaningful and sometimes dangerous patterns in the data, as demonstrated by intelligible models that revealed a counterintuitive association between a history of asthma and a lower predicted risk of pneumonia (Caruana et al., 2015). These standards motivate an analysis that compares a transparent ASA based baseline against flexible models, evaluates discrimination and calibration together, and prefers interpretable methods in order to support clinical trust.

2.7. Extracting functioning from clinical notes

A final area of literature concerns the extraction of structured information on patient functioning from free text clinical notes. Physiotherapy assessments and other clinical notes contain information on a patient's functioning that is not captured in the structured fields of a surgical registry, and natural language processing offers a way to make this information usable at scale. A proof of concept developed at Amsterdam UMC together with the Computational Lexicology and Terminology Lab at Vrije Universiteit Amsterdam trained classifiers to assign categories of the International Classification of Functioning, Disability and Health to sentences in Dutch electronic health record notes, reaching an average inter annotator agreement of 0.81 and F1 scores of around 0.7 for several functioning categories (Meskers et al., 2022). These F1 scores reflect moderate rather than high accuracy, and the classifiers were developed on general clinical notes rather than physiotherapy assessments specifically, so their outputs are treated here as a useful but imperfect signal rather than ground truth. These classifiers, together with the physiotherapy risk conclusion recorded during preoperative screening, are the basis of the fourth sub question. Instead of ignoring this area of research, it is examined directly, both as a single preoperative snapshot of functioning and as a trajectory over the weeks before surgery, to test whether functional information adds the signal that the structured registry variables may lack.

2.8. Summary and position of this study

The literature reviewed in this chapter supports two conclusions that together define the position of this study. Clinically, complications after oesophageal cancer surgery are frequent and consequential, affecting around 25 to 30 per cent of patients and measurably shortening survival, which makes their preoperative prediction a worthwhile goal. Methodologically, the tools currently available for this prediction do not transfer reliably from their development samples, are rarely assessed for calibration, and have not been shown to benefit consistently from added model complexity.

This defines two distinct gaps. The clinical gap is the absence of an accurate way to identify, before surgery, the patients most likely to suffer complications, which limits the targeting of preventive measures such as prehabilitation. The data science gap is the absence of a transparently reported and well calibrated model for these specific outcomes in this population, together with unresolved uncertainty about whether the operative course, the physiotherapy assessment, or flexible machine learning add anything beyond a simple and interpretable preoperative baseline. The present study addresses both gaps by developing and evaluating models for major complications, prolonged stay, and pulmonary complications on a large oesophageal cohort from the national registry under modern reporting and calibration standards, by testing in turn whether the operative course and the physiotherapy assessment raise the predictive ceiling, and by examining whether the factors the models rely on are consistent with established clinical knowledge.

3. Data and Methods

3.1. Study design and data sources

This study is a retrospective cohort study using prospectively collected data from the Dutch Upper Gastrointestinal Cancer Audit, the national surgical quality registry for oesophageal cancer surgery in the Netherlands, maintained by the Dutch Institute for Clinical Auditing (Voeten et al., 2021). The registry records preoperative patient characteristics, tumour and treatment details, the surgical procedure, and postoperative outcomes including complications, reinterventions, length of stay, readmission, and mortality (Busweiler et al., 2017). Data were obtained for patients treated at Amsterdam UMC between 2016 and 2025. Three data sources were used. The main registry table provides the structured preoperative variables, the operative details, and the recorded complications. An auxiliary table maintained alongside the registry at the participating centre provides additional preoperative measurements, namely pulmonary function and smoking history, and was merged onto the registry on the patient identifier. The physiotherapy assessment line of research uses information from two further sources, the preoperative physiotherapy notes processed by the ICF classifiers and a set of manual risk labels assigned by the supervising physiotherapist. These are described in Section 3.7. All sources were linked on a seven-digit zero padded patient identifier.

3.2. Cohort selection

The merge of the registry and the auxiliary table produced 1,201 records. Cohort selection then applied four steps. First, patients whose computed age at surgery fell outside the range of 18 to 100 years were removed, in each case because a placeholder date had been entered in place of a missing surgery or birth date. Second, where a patient had more than one surgical record, only the first surgery was retained as the index event, since later records represent reoperations or recurrences that carry a different risk profile. Third, salvage resections and prophylactic procedures were excluded on clinical grounds, as were records that appeared only in the auxiliary table without a corresponding surgical record and therefore had no demographic, staging, or outcome data. These three steps left 1,121 patients. Fourth, on the advice of the surgical team, the cohort was restricted to oesophageal tumours, because oesophageal and gastric resections differ substantially in their surgical risk and the small number of gastric and unclassified cases could not be pooled with the oesophageal patients without confounding the analysis. This left a final analytical cohort of 1,022 oesophageal patients. The 1,022 patients all had oesophageal tumours. The 62 gastric and 37 unclassified patients identified before this step were removed for the reasons given above. Restricting the cohort to a single tumour type, rather than pooling, follows the judgement of the surgical team that the operation is the dominant determinant of short-term complications and that the two operations are too different to model together. The physiotherapy assessment strand, which depends on linkage to the physiotherapy notes, is analysed on the same oesophageal patients.

3.3. Outcome derivation

Before any outcome was derived, registry sentinel codes that represent unknown or missing values, namely 9 for binary fields, 99 for the ASA and related scores, and 999 for measurements such as length of stay, were recoded as missing following the registry field options dictionary. Five binary outcomes were derived. The two headline outcomes are major complication, defined as a complication of Clavien-Dindo grade IIIb or higher, and prolonged length of stay. Pulmonary complication is the third principal outcome and the natural endpoint for the physiotherapy assessment strand, because the assessment is itself a judgement of pulmonary risk. Reintervention and a textbook outcome composite complete the set.

The Clavien-Dindo classification grades complications by the intervention required to treat them, and grade IIIb or higher corresponds to interventions requiring general anaesthesia, admission to intensive care, or resulting in death (Dindo et al., 2004). The highest grade recorded across all complication types was taken as each patient's overall severity, and the threshold of IIIb or higher was selected as a clinically meaningful definition of a major complication, consistent with the severity thresholds used in the textbook outcome literature for this surgery (Kalff et al., 2021). Complications were captured by the registry up to thirty days after discharge, and death and unscheduled readmission within this window both map to the highest Clavien-Dindo grade, so all major complications are captured. For the pulmonary and complication grade outcomes a clinical coding rule was applied to maximise the usable cohort: patients for whom no complication of any kind was recorded were coded as not having the complication, since the absence of any complication entails the absence of a pulmonary or graded one, while patients with a recorded complication but an unknown specific status were left as missing. Prolonged length of stay was defined as a postoperative stay longer than seven days, since the normal discharge window at this centre is day six to seven, so that a longer stay marks a departure from the expected course, and because the length of stay protocol has shortened over the study period, surgery year was examined as a possible adjustment. The textbook outcome was defined as the joint absence of a major complication, reintervention, readmission, and prolonged stay, together with discharge to home and survival to thirty days. Outcome frequencies are reported in Chapter 4.

3.4. Preoperative predictors

Candidate predictors were restricted to information available before surgery. The set was assembled from variables identified as clinically relevant by the surgical supervisors, preoperative factors established in the literature as predictors of complications after this surgery (Baranov et al., 2022; van Kooten et al., 2022), and the comorbidity and staging fields of the registry. A central design decision concerned the size of the predictor set relative to the number of events, since a rule of at least ten events per predictor is widely used to avoid overfitting (Peduzzi et al., 1996). The final preoperative set fixes thirteen clinically grounded predictors, applied uniformly to every outcome so that there is no outcome specific variable selection. They are listed in Table 3.1. With this set the number of events per predictor ranges from about ten for the major complication outcome to about twenty-seven for the textbook outcome.

Table 3.1. The thirteen preoperative predictors used in the surgical model.

Predictor	Definition	Type
Age at surgery	Years between birth date and surgery date	Continuous
Sex	Male versus female	Binary
Body mass index	Weight in kg divided by height in metres squared	Continuous
ASA score	American Society of Anesthesiologists physical status	Ordinal
Chronic pulmonary disease	Recorded chronic lung comorbidity	Binary
Charlson Comorbidity Index	Weighted comorbidity score built from registry components	Continuous
Prior abdominal surgery	Documented previous abdominal operation	Binary
Immunosuppression	Recorded immunosuppressive medication	Binary

Neoadjuvant therapy received	Any neoadjuvant treatment before surgery	Binary
Neoadjuvant chemoradiation	Neoadjuvant regimen was chemoradiation	Binary
Clinical T stage	Clinical tumour stage	Ordinal
Clinical N stage	Clinical nodal stage	Ordinal
Adenocarcinoma histology	Adenocarcinoma versus other histology	Binary

Two further candidate predictors were held out of the shared model. Smoking history was recorded for about seventy-three per cent of patients and adding it under complete case analysis removed roughly three hundred patients and pushed the rarer outcomes below ten events per predictor, so smoking was retained only for a sensitivity analysis. Preoperative weight loss in kilograms was likewise reserved for sensitivity analysis. Pulmonary function measurements were recorded for only a minority of patients and were examined in earlier exploratory work, where they added no discernible value, so they were not carried into the main analysis.

3.5. Operative and during stay variables

To test whether the operative course adds predictive value, four operative variables that are recorded for essentially every patient were defined: a minimally invasive versus open approach, a transthoracic versus other route of resection, a cervical versus other anastomosis location, and conversion from a minimally invasive to an open procedure. These four form the operative block that is added to the preoperative set in the third sub question. Two further operative measurements, operative duration and intraoperative blood loss, were recorded for about three quarters of patients and were reserved for a sensitivity analysis because of their missingness. Two during stay variables, days in intensive care and intensive care readmission, were retained for description but were not used as predictors of the complication outcomes, since they follow rather than precede the outcomes.

3.6. Comorbidity representation

The registry records a set of individual comorbidities, most of which are present in fewer than five per cent of patients and are therefore individually uninformative and unstable in a regression model. Rather than entering them separately, comorbidity burden was represented in two complementary ways. A Charlson Comorbidity Index was constructed from the registry components, following the standard weighting, and the ASA score was retained as a single summary of overall fitness. The relationship between the two is examined as a sensitivity analysis. The decision to summarise comorbidity in this way, rather than as a list of individual conditions, follows the registry study that found the ASA score, and not a detailed comorbidity count, to be the relevant preoperative risk factor for this surgery (Baranov et al., 2022).

3.7. Physiotherapy assessment and ICF features

The note corpus comprises 504,964 clinical notes for 1,862 unique patients, recorded by all healthcare professionals across the full pre and postoperative course rather than by physiotherapists alone. Two kinds of feature were derived. The first is a binary risk conclusion, low or high, taken from the preoperative physiotherapy assessment. Conclusions were extracted from the structured conclusion field of physiotherapy assessment notes by a fixed rule that locates the concluding statement and classifies its risk wording, with an ordered keyword scheme distinguishing low risk from the several gradations of raised risk. A conclusion was assigned to each note as of its date and propagated forward, so that the conclusion current before surgery is used and no information from after surgery enters the feature. Where the notes carried no usable conclusion for a patient, the manual risk label assigned by the supervising physiotherapist was used as a fallback. This yielded a conclusion for 1,137 patients, of whom 440 are in the oesophageal surgical cohort.

The second kind of feature is the ICF coded functioning itself. For each of six functioning domains that are well populated before surgery, exercise tolerance, respiration, weight maintenance, eating, energy and drive, and walking, two preoperative summaries were computed, the value of the note closest to surgery and the worst value recorded before surgery. To test whether the way functioning changes before surgery carries information that a single value does not, two trajectory features were added per domain, the change from the earliest to the latest preoperative value and the linear slope of the score against time toward surgery, the latter computed only for patients with at least three preoperative measurements. All assessment features use only notes dated before the surgery date. The distribution of each ICF category before surgery, and the preoperative course of the six well populated domains over the weeks before surgery, are shown in Appendix D.

3.8. Missingness handling

A complete case analysis was adopted throughout, with no imputation. Imputation was avoided because the missingness mechanism in the registry cannot be verified, and a complete case analysis keeps the path from raw data to model transparent. Each model was fitted on the patients with complete data for the predictors and the outcome in question, and any candidate predictor missing for more than half of the cohort was removed before this selection. Of the 1,022 oesophageal patients, 929 had complete data on all thirteen preoperative predictors, and after also requiring the outcome the surgical models for the principal outcomes were fitted on about 860 patients. To check that excluding incomplete records did not introduce obvious selection bias, the 929 retained patients were compared with the 93 excluded patients on age, sex, ASA, and the principal outcome rates (Table 3.2). The groups were broadly similar, with all standardised differences below 0.2, but the excluded patients were modestly sicker, with a higher proportion at ASA III or above (37.6 versus 31.5 per cent) and a higher rate of major complications (21.5 versus 14.5 per cent). The complete case cohort therefore slightly underrepresents the highest risk patients. Because every model comparison in this study is made on the same complete case sample, this selection affects the generalisability of the absolute risk estimates rather than the internal comparisons of added value, and it is revisited in Section 5.3.

Table 3.2. Retained versus excluded patients.

Characteristic	Retained (n = 929)	Excluded (n = 93)	Standardized Mean Difference
Age, mean (SD)	65.8 (9.1)	66.9 (8.5)	-0.12
Male, %	79.2	72.0	0.17
ASA III or above, %	31.5	37.6	-0.13
Major complication, %	14.5	21.5	-0.18
Pulmonary complication, %	23.9	25.8	-0.04
Prolonged stay, %	69.0	64.5	0.09

3.9. Statistical analysis

Each sub question was translated into a binary classification task. The primary model was logistic regression with an L2 penalty. The strength of the penalty was tuned once per feature set by selecting the value that minimised the cross validated log loss, and that value was then fixed for the evaluation, so that the penalty was set without reference to the held-out discrimination. Continuous predictors were standardised within each cross-validation fold to prevent leakage from the held-out data. Class imbalance was handled only through the penalty and the choice of metric, not through resampling or class weighting, because resampling has been shown to harm calibration without improving discrimination (van den Goorbergh et al., 2022).

Performance was estimated by stratified 5-fold cross validation, with predicted probabilities from the held-out folds used to compute every metric. Discrimination was measured by the area under the receiver operating characteristic curve, with a 95 per cent confidence interval obtained from bootstrap resamples of the held-out predictions. Calibration was summarised by the calibration slope and the Brier score (Van Calster et al., 2019) shown graphically as calibration curves. Clinical utility was assessed by decision curve analysis (Vickers & Elkin, 2006), which reports the net benefit of using the model to guide treatment across a range of decision thresholds, relative to treating all patients or none. For the textbook outcome the model was specified to predict failure of the outcome, so that the decision curve reads in the same direction as the complication outcomes. For SQ1, an ASA only baseline and the thirteen variable preoperative model were compared for each outcome.

For SQ2, the factors carrying the signal were examined through unadjusted univariable odds ratios, computed by fitting a single predictor logistic regression and deriving the Wald confidence interval analytically, and through the adjusted odds ratios of the full model, shown as forest plots for the headline outcomes. For SQ3, the preoperative model was compared with a model that adds the four operative variables, and the difference in the area under the curve was assessed by a paired bootstrap, from which a confidence interval for the difference and the bootstrap probability that the larger set was superior were obtained. To answer the question of model complexity, the tuned logistic model was compared on the same features against a lasso penalised model and a random forest. Heavier gradient boosting models could not be evaluated in the secure environment because they exhausted its resources, and the comparison is therefore confined to the three models that ran reliably. The random forest nonetheless provides a flexible, non-linear benchmark, and as it did not outperform the logistic model, consistent with evidence that machine learning rarely improves on regression for clinical prediction in studies at low risk of bias (Christodoulou et al., 2019), gradient boosting was unlikely to change this conclusion. For SQ4, the physiotherapy conclusion and each ICF feature were added in turn to the surgical model on their own complete case sample, and the change in the area under the curve was reported.

3.10. Sensitivity analyses

Four sensitivity analyses were specified. The first examines the relationship between the ASA score and the Charlson Comorbidity Index, to test whether the ASA score can be read as a summary of comorbidity burden. The second adds smoking history and preoperative weight loss to the preoperative model on the subset of patients for whom they were recorded, to confirm that holding them out did not discard signal. The third adds operative duration and blood loss to the operative model on the subset for whom they were recorded.

The fourth compares the variables selected by a random forest, through permutation importance, with those retained by the lasso, to check whether any predictor carries a non-linear signal that the logistic model would miss.

3.11. Software and computational environment

All analyses were carried out in Python, using the pandas library for data handling and the scikit learn library for the logistic regression, the lasso, and the random forest. The analysis was implemented as a sequence of numbered notebooks, beginning with a single preprocessing notebook that produces one cleaned cohort file used by every subsequent step, so that the path from the raw registry extract through cohort selection, variable preparation, modelling, and the physiotherapy assessment strand can be reproduced in order. The transformations described in this chapter, the recoding of sentinel values, the construction of the Charlson index and the operative variables, the derivation of the outcomes, and the extraction of the physiotherapy conclusions and ICF features, were implemented in code and applied uniformly to all patients. The principal data transformations in this chapter are reproduced as annotated code in Appendix B. The reporting follows the TRIPOD plus AI guideline for prediction models (Collins et al., 2024).

3.12. Ethical and data governance considerations

All modelling of patient data was carried out within a secure virtual research environment, and only aggregate results and figures left it. Access was restricted and logged, and the registry data fell under a non-WMO waiver from the Medical Ethics Committee (METc number 2020.277) and the opt out consent model applicable under the General Data Protection Regulation. Patients are described throughout this thesis only in aggregate, and no result is reported at a level of detail that could identify an individual. The use of generative artificial intelligence in the project is described in Appendix A. The responsible interpretation of the models, including the risk that weak discrimination could give false reassurance, is considered in the discussion.

4. Results

4.1. Cohort description

The analytical cohort comprised 1,022 patients with oesophageal tumours treated at Amsterdam UMC between 2016 and 2025. Prolonged stay was defined as a postoperative stay longer than seven days. The cohort had a median age of 67 years, was 79 per cent male, and had a median Charlson Comorbidity Index of one. Most patients were ASA class 2 or 3, and 95 per cent received neoadjuvant therapy, so that both variables vary only modestly across the cohort. Table 4.1 reports the frequency of each outcome.

Table 4.1. Outcome frequencies in the full cohort.

Outcome	Events / known	Prevalence
Any complication	588 / 1,022	57.5%
Clavien-Dindo II or higher	481 / 1,017	47.3%
Major complication (Clavien-Dindo IIIb or higher)	154 / 1,017	15.1%
Pulmonary complication	246 / 1,022	24.1%
Reintervention	234 / 1,022	22.9%
Prolonged stay (over 7 days)	700 / 1,021	68.6%
Readmission within 30 days	150 / 1,008	14.9%
Textbook outcome	598 / 1,002	59.7%
Mortality within 90 days	30 / 1,022	2.9%

Table 4.2 describes the cohort by the major complication outcome. Patients who suffered a major complication had a higher Charlson index, were more often ASA class 3 or higher, more often had chronic pulmonary disease, less often received neoadjuvant therapy, and more often had a cervical anastomosis. Age, body mass index, and sex did not differ. These contrasts preview the drivers examined under the second sub question.

Table 4.2. Selected characteristics by major complication. Continuous variables are median with interquartile range, binary variables are count and per cent.

Characteristic	No major (n=863)	Major (n=154)	p
Charlson index	0.0 [0.0, 2.0]	1.0 [0.0, 2.0]	0.002
ASA class 3 or higher	254 (29.4%)	72 (46.8%)	<0.001
Chronic pulmonary disease	80 (9.3%)	27 (17.5%)	0.003
Neoadjuvant therapy	825 (95.6%)	137 (89.0%)	0.002
Cervical anastomosis	172 (19.9%)	53 (34.4%)	<0.001
Adenocarcinoma	706 (81.8%)	113 (73.4%)	0.02
Age, years	67.6 [60.6, 72.4]	67.2 [61.2, 72.9]	0.547
Male	681 (78.9%)	119 (77.3%)	0.726

4.2. Preoperative prediction of the principal outcomes (SQ1)

Discrimination from the preoperative record was modest for every outcome. The thirteen variable preoperative model reached a cross validated area under the curve of 0.594 for major complications, 0.585 for prolonged stay, 0.548 for pulmonary complications, 0.543 for reintervention, and 0.597 for the textbook outcome, on complete case samples of about 860 patients. The preoperative model improved on the ASA only baseline most clearly for prolonged stay and reintervention, where the ASA score alone discriminated barely above chance. For major complications and pulmonary complications the ASA score already carried almost all of the available signal, reaching 0.586 and 0.541 on its own. Table 4.3 reports the discrimination of the baseline and the preoperative model, and Figure 4.1 compares the ASA baseline, the preoperative, and the perioperative models across the outcomes.

Table 4.3. Discrimination of the ASA baseline and the preoperative model, cross validated area under the curve.

Outcome	N	Events	ASA only	Preoperative
Major complication	858	126	0.586	0.594
Prolonged stay	862	602	0.518	0.585
Pulmonary	863	212	0.541	0.548
Reintervention	863	195	0.507	0.543
Textbook outcome	845	345	0.566	0.597

Calibration was assessed for every model alongside discrimination and is reported with the perioperative models in Section 4.4, since the preoperative and perioperative models perform too similarly to separate. Across the principal outcomes the events per predictor for the preoperative model ranged from about ten for major complications to about twenty-seven for the textbook outcome, so the modest discrimination is not a consequence of too few events for the number of predictors.

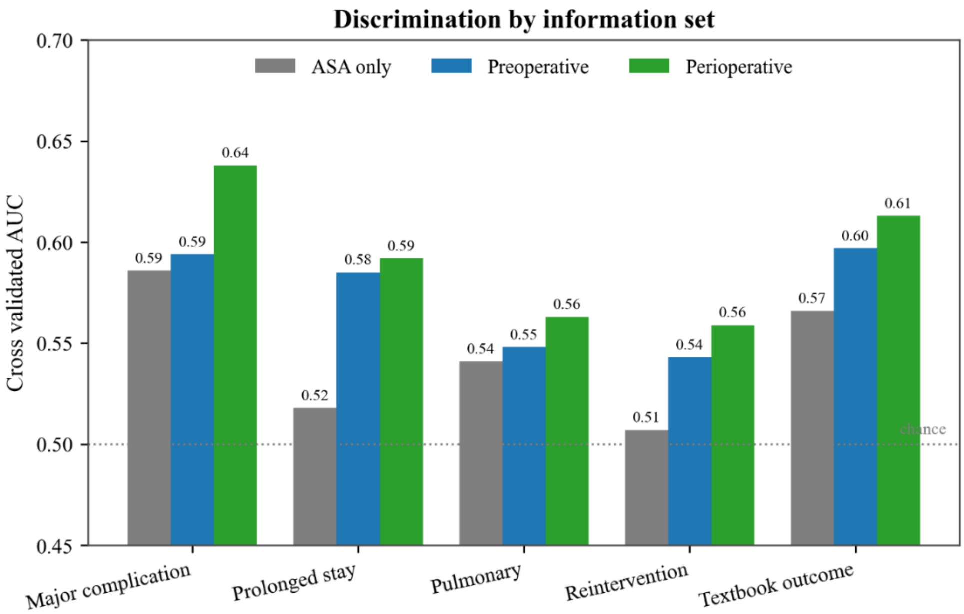

Figure 4.1. Cross validated discrimination for the ASA baseline, the preoperative model, and the perioperative model, by outcome.

4.3. Factors carrying the signal (SQ2)

The univariable screen identified a small and clinically coherent group of variables that raise risk across the outcomes. The strongest preoperative associations were chronic pulmonary disease, with odds ratios of about 2.1 for major complications and 2.6 for prolonged stay, the ASA score, with an odds ratio of 1.89 for major complications, and the Charlson index. Among operative variables, a cervical anastomosis was associated with raised risk of most outcomes. Neoadjuvant therapy was associated with lower risk of complications and with a higher chance of a textbook outcome, and adenocarcinoma histology was associated with lower pulmonary risk.

In the adjusted models for the two headline outcomes the operative variables again emerged among the strongest factors. Conversion from a minimally invasive to an open procedure carried the largest adjusted odds ratio for major complications, at 3.44 with a 95 per cent confidence interval from 1.35 to 8.78, and pointed the same way for prolonged stay, at 2.26, although conversion was rare and that interval was wide and included one. For prolonged stay the estimable factors whose intervals excluded one were chronic pulmonary disease, at 1.93 with an interval from 1.00 to 3.73, and the Charlson index, at 1.29 with an interval from 1.09 to 1.53. A cervical anastomosis and a higher ASA score raised the odds of major complications, while neoadjuvant therapy, a transthoracic resection, and adenocarcinoma histology were protective. Age, sex, and body mass index carried adjusted odds ratios close to one. Table 4.4 reports the adjusted odds ratios of the strongest factors for the two headline outcomes, and the full forest plots are shown in Figures 4.2 and 4.3.

Table 4.4. Adjusted odds ratios of the strongest factors in the headline models, with 95 per cent confidence intervals where estimable.

Factor	**Major complication**	**Prolonged stay**
Conversion to open	3.44 (1.35, 8.78)	2.26 (0.72, 7.15)
Cervical anastomosis	1.63	1.31
Chronic pulmonary disease	1.39 (0.74, 2.63)	1.93 (1.00, 3.73)
Charlson index	1.07 (0.88, 1.30)	1.29 (1.09, 1.53)
Neoadjuvant therapy	0.49 (0.17, 1.35)	0.44 (0.11, 1.72)
Adenocarcinoma	0.62	0.93 (0.61, 1.40)

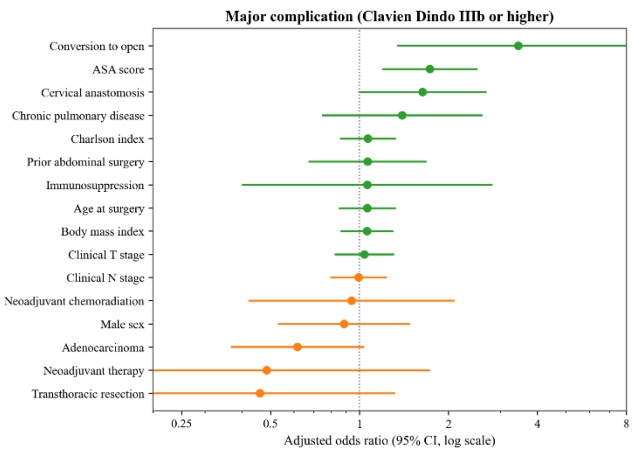

Figure 4.2. Adjusted odds ratios for major complications.

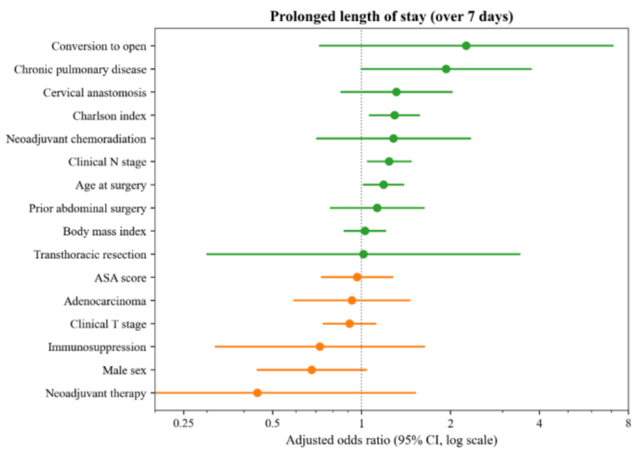

Figure 4.3. Adjusted odds ratios for prolonged stay.

4.4. Added value of the operative course (SQ3)

Adding the four operative variables to the preoperative model produced a clear improvement for major complications and almost none for the others. The area under the curve rose from 0.594 to 0.638 for major complications, a gain of 0.044 that the paired bootstrap favoured with a probability of 0.995, whereas for prolonged stay it rose only from 0.585 to 0.592, a gain of 0.007. For pulmonary complications, reintervention, and the textbook outcome the operative block added about 0.016 each. Major complications was therefore the only outcome for which the operative course produced a substantial improvement. Calibration of the perioperative models was good for the textbook outcome, with a calibration slope of 0.99, moderate for major complications and prolonged stay, at 0.81 and 0.82, and poor for pulmonary complications, at 0.48, where the model was overconfident (Figure 4.4). Decision curve analysis showed a positive net benefit over treating all or none for major complications, the textbook outcome, and reintervention across the clinically relevant threshold range, but no net benefit at any threshold for prolonged stay. Table 4.5 reports the comparison.

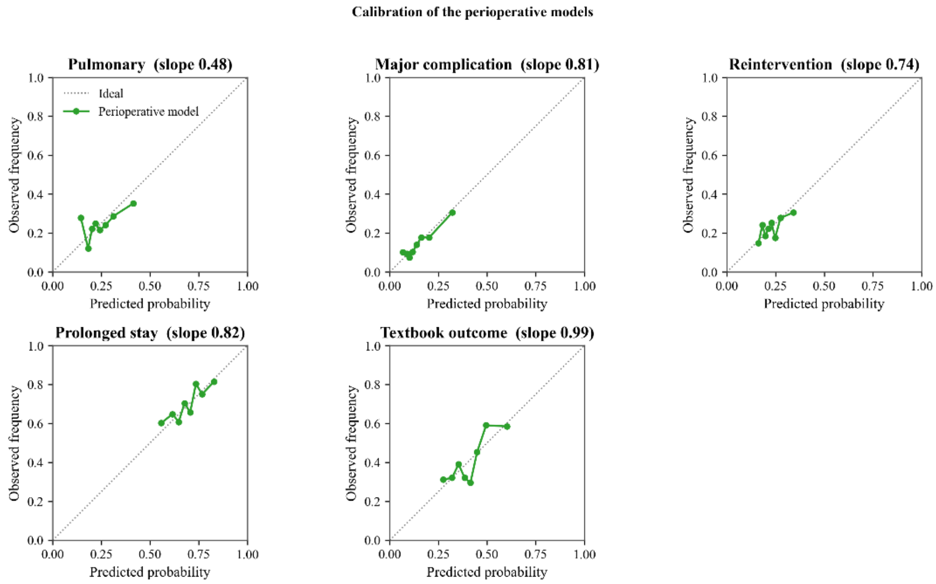

Figure 4.4. Calibration of the perioperative models. Each panel plots observed against predicted risk, with the calibration slope in the title. The dotted line marks perfect calibration; points below it indicate overprediction.

Table 4.5. Preoperative versus perioperative discrimination, with the calibration slope of the perioperative model.

Outcome	**Preoperative**	**Perioperative**	**Gain**	**P(perioperative better)**	**Calibration slope**
Major complication	0.594	0.638	+0.044	0.995	0.81
Prolonged stay	0.585	0.592	+0.007	0.895	0.82
Pulmonary	0.548	0.563	+0.016	0.91	0.48
Reintervention	0.543	0.559	+0.016	0.935	0.74
Textbook outcome	0.597	0.613	+0.016	0.98	0.99

Replacing the logistic model with more flexible methods did not improve on it. On the perioperative features a random forest reached 0.643 for major complications against 0.638 for the tuned logistic model, and 0.580 for pulmonary complications against 0.563, small differences in both directions, while for prolonged stay the logistic model was better at 0.592 against 0.560. The lasso performed almost identically to the logistic model throughout. Table 4.6 and Figure 4.5 report the panel. Adding operative duration and blood loss to the operative model, in the sensitivity analysis, did not improve discrimination for any outcome and reduced it for major complications, so these two measurements add no usable signal beyond the four operative variables already included.

Table 4.6. Model family comparison on the perioperative features, cross validated area under the curve.

Outcome	Logistic (tuned)	Lasso	Random forest
Major complication	0.638	0.636	0.643
Prolonged stay	0.592	0.589	0.560
Pulmonary	0.563	0.548	0.580
Reintervention	0.559	0.543	0.575
Textbook outcome	0.613	0.612	0.615

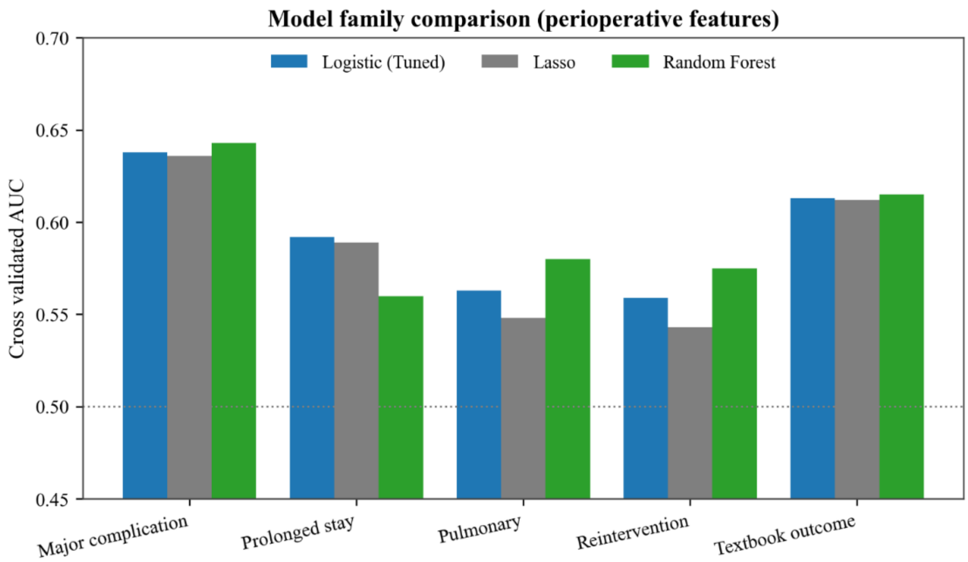

Figure 4.5. Discrimination of the logistic model, the lasso, and the random forest on the perioperative features.

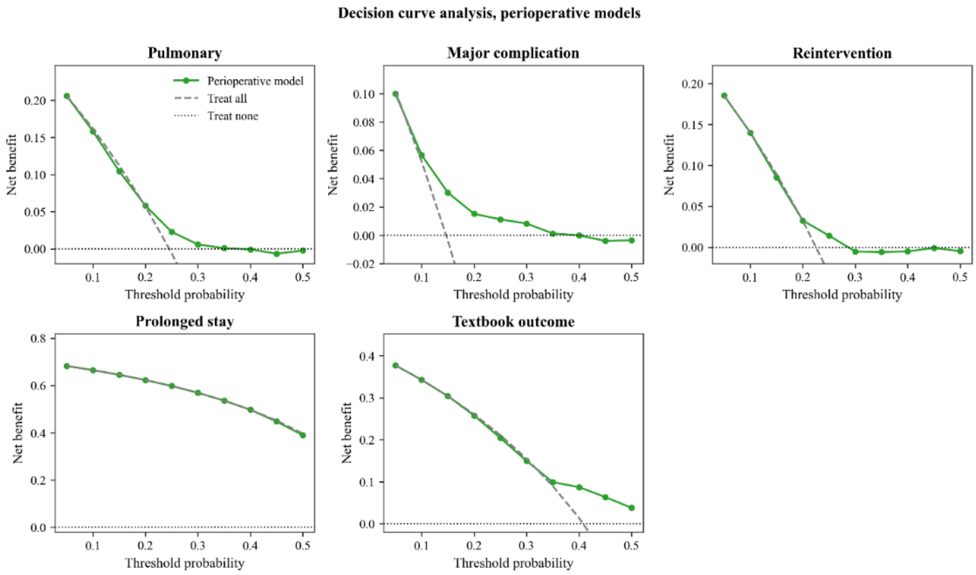

Figure 4.6. Decision curves for the perioperative models, showing net benefit against treating all or none.

4.5. Added value of the physiotherapy assessment (SQ4)

The physiotherapy strand linked 516 oesophageal patients to preoperative physiotherapy notes and 440 to a usable risk conclusion. This is about half of the cohort, since not every patient passed through the preoperative screening pathway or had physiotherapy notes available in the linked records, so the assessed subset may not be fully representative.

The conclusion tracked risk in the expected direction. Among patients with a high conclusion the pulmonary complication rate was 28.0 per cent against 22.0 per cent for a low conclusion, the major complication rate was 17.6 against 12.1 per cent, and the prolonged stay rate was 71.0 against 52.8 per cent. The conclusion on its own, however, discriminated weakly, with areas under the curve between 0.52 and 0.57, and adding it to the surgical model did not improve discrimination for any outcome. The gains were negative for pulmonary and major complications, with the confidence interval for major complications lying entirely below zero, and negligible for prolonged stay. Because this strand is restricted to the patients with a linked physiotherapy assessment, the surgical baselines in Table 4.7 were refitted on that smaller subsample, which is why they differ from the whole-cohort figures in Tables 4.3 and 4.5. Table 4.7 reports the comparison.

Table 4.7. Physiotherapy conclusion: outcome rate by conclusion and added value over the surgical model.

Outcome	Rate low	Rate high	Conclusion alone	Surgical	Surgical + conclusion	Gain
Pulmonary	22.0%	28.0%	0.518	0.590	0.537	-0.053
Major complication	12.1%	17.6%	0.535	0.557	0.506	-0.052
Prolonged stay	52.8%	71.0%	0.572	0.614	0.621	+0.008

The ICF coded functioning gave the same answer. Adding each preoperative functioning score to the surgical model produced changes in the area under the curve that were near zero or negative, and the few small positive changes, the largest about 0.02, did not persist across feature definitions. Exercise tolerance, the domain most directly related to fitness, added nothing for any outcome.

The trajectory analysis was viable, since patients carried a median of five to fourteen preoperative measurements per domain, but it returned the same result, with the largest gain about 0.016 sitting on a baseline that carried no signal. A validation check confirmed that exercise tolerance differed clearly between the low and high conclusions, being lower in high-risk patients, with a Mann-Whitney p value below 0.001, and that it declined toward surgery in high-risk patients while remaining stable in low-risk ones, with a p value of 0.015. The classifier therefore agrees with the assessment even though neither adds to short term prediction. The clinical team confirmed that the classifier codes a higher exercise tolerance score as better functioning, so the lower scores observed in high-risk patients correctly reflect worse functioning. Figures 4.7 and 4.8 summarise the physiotherapy assessment.

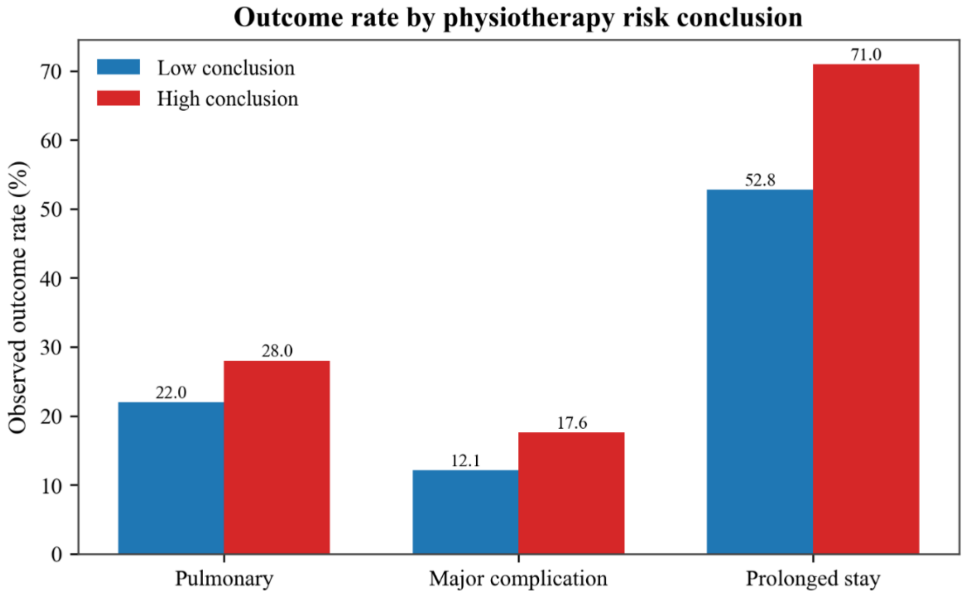

Figure 4.7. Outcome rate by physiotherapy conclusion.

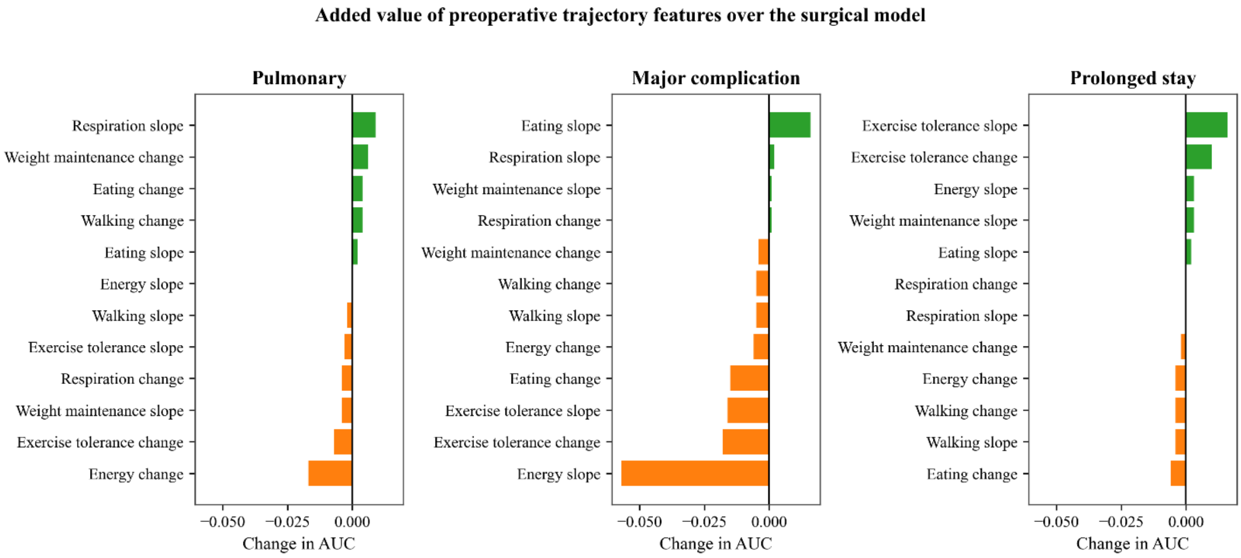

Figure 4.8. Change in discrimination when a trajectory feature is added to the surgical model. The full preoperative ICF profile across all functioning domains, split by the low and high conclusion, is shown in Appendix C and follows the same pattern, with the high-risk group scoring lower on most domains including exercise tolerance.

4.6. Sensitivity analyses

The ASA score behaved as a summary of comorbidity burden. Mean Charlson index rose from 0.36 at ASA class 1 to 0.74 at class 2 and 1.55 at class 3, and the prevalence of chronic pulmonary disease rose from zero to about 7 and 20 per cent across the same classes, with a Spearman correlation between ASA and Charlson of 0.35 (Figure 4.9). This supports treating the ASA score as a compact representation of comorbidity, consistent with the registry literature (Baranov et al., 2022). Adding smoking history and preoperative weight loss to the preoperative model changed discrimination by at most 0.02 for the principal outcomes, confirming that holding them out did not discard signal. The reverse engineering comparison found that the random forest favoured the continuous variables, age and body mass index, a known property of permutation importance, while the lasso retained the operative flags, and the variables recurring across both methods, conversion, the cervical anastomosis, the ASA score, and chronic pulmonary disease, are the same drivers identified in the adjusted models, with no sign of a hidden non-linear signal.

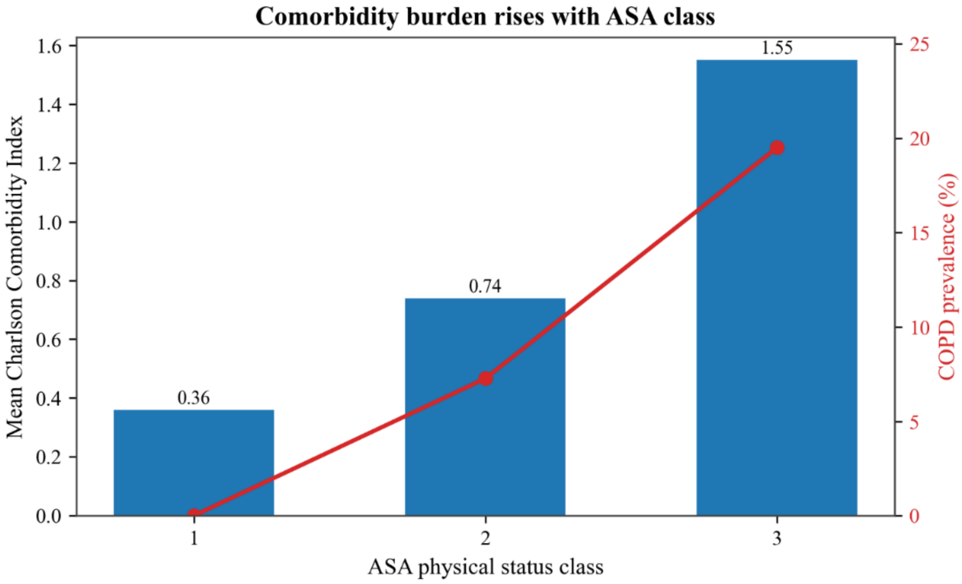

Figure 4.9. Mean Charlson Comorbidity Index by ASA class.

5. Discussion

5.1. Principal findings

This study asked how far complications after oesophageal cancer surgery can be predicted from routinely available information, and whether the operative course or the preoperative physiotherapy assessment raises that ceiling. The answer is consistent across every layer examined. The structured preoperative record predicts the principal outcomes only modestly, with cross validated areas under the curve between 0.55 and 0.60, and the richer preoperative model improves on the ASA score for most outcomes but does not lift discrimination into a clinically useful range. The factors that carry the signal are clinically coherent, with chronic pulmonary disease, the ASA score, the Charlson index, and, once the operative course is included, conversion to an open procedure and a cervical anastomosis emerging as the consistent drivers. The operative course adds a clear improvement for major complications, the only outcome for which it does so, and almost nothing for the others, including prolonged stay. The physiotherapy assessment, both the risk conclusion and the ICF coded functioning, tracks risk in the expected direction but adds no discrimination beyond the surgical model, whether the functioning is taken as a single preoperative snapshot or as a trajectory over the weeks before surgery. Flexible machine learning does not improve on the interpretable logistic model. The ceiling on prediction is therefore reached early, and none of the additional layers of routine information moves it appreciably.

Two features of the analysis give these findings weight. The result is stable under proper cross validation, with calibration reported alongside discrimination rather than discrimination alone, and the headline outcomes were modelled with about ten or more events per predictor, so the modest performance is not an artefact of an inadequate validation procedure or of too few events. The consistency of the answer across four independent angles, the preoperative record, the operative course, the assessment conclusion, and the ICF scores tested two ways, is itself the strongest evidence that the ceiling is a property of the information available rather than of any single modelling choice.

5.2. Interpretation in light of existing literature

These findings align closely with the existing literature rather than contradicting it. The early conclusion that there are no good preoperative predictors of pulmonary complications after oesophagectomy (Avendano et al., 2002) is reproduced here on a much larger and more recent cohort with modern methods. The registry evidence that the ASA score, rather than a detailed comorbidity index, carries the available preoperative signal (Baranov et al., 2022) is supported both by the modest performance of the preoperative model and by the sensitivity analysis showing the ASA score to be a compact summary of comorbidity. The same registry study identified the surgical approach as an independent risk factor, and the present finding that the operative course adds real value for major complications, driven by conversion and the anastomosis, is consistent with that. The absence of any advantage for flexible machine learning matches the systematic review evidence that, among studies at low risk of bias, there is no performance benefit of machine learning over logistic regression for clinical prediction (Christodoulou et al., 2019). The recurrence of the same limited set of predictors across many studies (van Kooten et al., 2022) had already suggested that the preoperative signal is real but limited and well characterised, and the present study confirms this directly and quantitatively, while extending it to the operative and functional layers that earlier work did not test together.

The result on the physiotherapy assessment deserves a specific interpretation. That the conclusion tracks risk yet adds nothing to the surgical model is not a failure of the assessment but a consequence of redundancy, since the assessment is built from fitness and comorbidity information, the ASA score, chronic pulmonary disease, and age, that the surgical model already contains. The natural language classifier supports this reading, because the exercise tolerance it extracts agrees with the physiotherapist's conclusion and declines toward surgery in the patients judged high risk. This agreement should be read with care, since the classifier extracts functioning from the same notes that record the physiotherapist's conclusion, so the validation is not fully independent. Two further points temper the negative result. The assessment is not a passive measurement but an intervention, since a high-risk conclusion can change care through exercise advice or referral to further physiotherapy, which may weaken its association with later complications. And a physical assessment is plausibly better suited to predicting longer term functional recovery than the short-term surgical complications studied here, which are dominated by the operation itself. The ICF coded functioning, although a richer, independent measurement, likewise did not improve prediction of these short-term outcomes.

5.3. Strengths and limitations

The study has several strengths. It uses a large, routinely collected, and previously unanalysed registry cohort from a national audit, with clinically meaningful outcomes graded by standardised definitions. It follows a modern reporting standard, evaluates discrimination and calibration together, assesses clinical utility through decision curve analysis, and avoids class imbalance corrections known to harm calibration (van den Goorbergh et al., 2022). A parsimonious predictor set sized to the events, with the operative and functional layers tested over it, makes the negative result credible rather than a consequence of an underpowered model. Linking the physiotherapy assessment and the ICF classifier output to surgical outcomes, and testing functioning as both a snapshot and a trajectory, extends the analysis beyond the structured registry alone.

The limitations are equally important. The analysis uses a single centre extract, which limits external validity, and a complete case approach. As shown in Section 3.8, the excluded patients were modestly sicker than those retained, with more at ASA III or above and a higher rate of major complications, so the models were developed on a marginally healthier sample. Because every comparison was made on the same complete case sample, this affects the generalisability of the absolute risk estimates rather than the internal comparisons of added value that answer the research questions. For major complications, adding the operative variables left about eight events per predictor, so that perioperative estimate should be read with caution. Conversion to an open procedure, the strongest single factor, was rare, with a correspondingly wide confidence interval, and because conversion, the minimally invasive approach, and the resection route describe the same operation, the unstable estimate for the minimally invasive approach is omitted from the forest plots. Pulmonary function and smoking were dropped for missingness, though the sensitivity analyses showed neither added value where recorded. The outcomes depend on registry coding, and the rule used to maximise the cohort assumes that patients with no recorded complication genuinely had none, so any under-recording would bias the counts. Finally, for most linked patients the physiotherapy conclusion was the manually assigned label rather than one extracted automatically; these manual labels are the reference standard used in routine care, so the finding speaks reliably to the assessment conclusion itself, while the automated extraction contributed conclusions for fewer patients.

5.4. Ethical implications

The central ethical implication follows directly from the weak performance. A model that discriminates only weakly, and that is in places overconfident, must not be presented or deployed as a preoperative risk tool, because it would give false reassurance for some patients and unwarranted alarm for others, and could distort the targeting of prehabilitation rather than improve it. Subgroup performance, for example by age, sex, or tumour stage, was not assessed here, and a formal fairness evaluation across such groups would be a prerequisite before any model of this kind could be considered for deployment. The responsible conclusion is that these routinely collected variables are not sufficient on their own to support individual risk prediction for these outcomes, and that the honest reporting of this fact, with calibration and uncertainty made explicit, is itself the useful contribution. All modelling was conducted within a secure environment under the applicable medical ethics and data protection framework, no patient level data left that environment, and results are reported only in aggregate, so that the privacy of the patients in the registry was protected throughout. The use of generative artificial intelligence in the project is documented in Appendix A.

5.5. Future work

Several directions follow from these findings. The first is external validation on data from other centres in the national audit, which would establish whether the modest ceiling found here generalises beyond a single centre. The second concerns the operative course: since conversion and the anastomosis carried the strongest signal, a prospective study designed around these variables, with enough events to estimate rare factors precisely, could clarify how much of the outcome is determined at the time of surgery rather than before it. The third concerns functioning. The physiotherapy assessment and the ICF classifiers did not add value here, but the analysis was limited to the patients who could be linked and to functioning measured at the preoperative stage. A larger linked cohort, assembled prospectively, paired with longer-term functional recovery endpoints rather than the thirty-day surgical complications studied here, and with functioning assessed across the full prehabilitation window rather than only at screening, would give a fairer test of whether functional status can contribute the signal that the structured record lacks. Given the consistent finding that routine information carries little signal for these short-term outcomes, the most promising approach is a richer and prospectively collected measure of patient functioning rather than a more complex model applied to the same variables.

5.6. Conclusion

This study developed and evaluated prediction models for major complications, prolonged stay, and pulmonary complications after oesophageal cancer surgery on a large registry cohort and tested in turn whether the operative course and the preoperative physiotherapy assessment raise the predictive ceiling. Routinely collected preoperative variables were found to carry only modest signal for these outcomes, the operative course added a clear improvement for major complications but not for the other outcomes, the physiotherapy conclusion and the ICF coded functioning tracked risk but added nothing beyond the surgical model, and flexible machine learning did not improve on a transparent baseline. These findings are consistent with the wider prediction modelling literature and indicate that accurate preoperative risk prediction for these outcomes will require information beyond what the current sources provide. The contribution of the study is a transparent, well validated, and honestly reported account of how far routine information can be taken for this clinical problem across four distinct layers.

6. References

Avendano, C. E., Flume, P. A., Silvestri, G. A., King, L. B., & Reed, C. E. (2002). Pulmonary complications after esophagectomy. The Annals of Thoracic Surgery, 73(3), 922-926. https://doi.org/10.1016/S0003-4975(01)03584-6

Baranov, N., Claassen, L., van Workum, F., & Rosman, C. (2022). Age and Charlson Comorbidity Index score are not independent risk factors for severe complications after curative esophagectomy for esophageal cancer: A Dutch population-based cohort study. Surgical Oncology, 43, 101789. https://doi.org/10.1016/j.suronc.2022.101789

Bona, D., Manara, M., Bonitta, G., Guerrazzi, G., Guraj, J., Lombardo, F., Biondi, A., Cavalli, M., Bruni, P. G., Campanelli, G., Bonavina, L., & Aiolfi, A. (2024). Long-term impact of severe postoperative complications after esophagectomy for cancer: Individual patient data meta-analysis. Cancers, 16(8), 1468. https://doi.org/10.3390/cancers16081468

Booka, E., Takeuchi, H., Suda, K., Fukuda, K., Nakamura, R., Wada, N., Kawakubo, H., & Kitagawa, Y. (2018). Meta-analysis of the impact of postoperative complications on survival after oesophagectomy for cancer. BJS Open, 2(5), 276-284. https://doi.org/10.1002/bjs5.64

Busweiler, L. A. D., Schouwenburg, M. G., van Berge Henegouwen, M. I., Kolfschoten, N. E., de Jong, P. C., Rozema, T., Wijnhoven, B. P. L., van Hillegersberg, R., Wouters, M. W. J. M., & van Sandick, J. W. (2017). Textbook outcome as a composite measure in oesophagogastric cancer surgery. British Journal of Surgery, 104(6), 742-750. https://doi.org/10.1002/bjs.10486

Caruana, R., Lou, Y., Gehrke, J., Koch, P., Sturm, M., & Elhadad, N. (2015). Intelligible models for healthcare: Predicting pneumonia risk and hospital 30-day readmission. Proceedings of the 21st ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 1721-1730. https://doi.org/10.1145/2783258.2788613

Christodoulou, E., Ma, J., Collins, G. S., Steyerberg, E. W., Verbakel, J. Y., & Van Calster, B. (2019). A systematic review shows no performance benefit of machine learning over logistic regression for clinical prediction models. Journal of Clinical Epidemiology, 110, 12-22. https://doi.org/10.1016/j.jclinepi.2019.02.004

Collins, G. S., Moons, K. G. M., Dhiman, P., Riley, R. D., Beam, A. L., Van Calster, B., Ghassemi, M., Liu, X., Reitsma, J. B., van Smeden, M., ... Logullo, P. (2024). TRIPOD+AI statement: Updated guidance for reporting clinical prediction models that use regression or machine learning methods. BMJ, 385, e078378. https://doi.org/10.1136/bmj-2023-078378

Dindo, D., Demartines, N., & Clavien, P. A. (2004). Classification of surgical complications: A new proposal with evaluation in a cohort of 6336 patients and results of a survey. Annals of Surgery, 240(2), 205-213. https://doi.org/10.1097/01.sla.0000133083.54934.ae

Kalff, M. C., Vesseur, I., Eshuis, W. J., Heineman, D. J., Daams, F., van der Peet, D. L., van Berge Henegouwen, M. I., & Gisbertz, S. S. (2021). The association of textbook outcome and long-term survival after esophagectomy for esophageal cancer. The Annals of Thoracic Surgery, 112(4), 1134-1141. https://doi.org/10.1016/j.athoracsur.2020.09.035

Low, D. E., Alderson, D., Cecconello, I., Chang, A. C., Darling, G. E., D'Journo, X. B., Griffin, S. M., Hölscher, A. H., Hofstetter, W. L., Jobe, B. A., Kitagawa, Y., Kucharczuk, J. C., Law, S. Y. K., Lerut, T. E., Maynard, N., Pera, M., Peters, J. H., Pramesh, C. S., Reynolds, J. V., ... van Lanschot, J. J. B. (2015). International consensus on standardization of data collection for complications associated with esophagectomy: Esophagectomy Complications Consensus Group (ECCG). Annals of Surgery, 262(2), 286-294. https://doi.org/10.1097/SLA.0000000000001098

Lu, S., Yan, M., Li, C., Yan, C., Zhu, Z., & Lu, W. (2019). Machine-learning-assisted prediction of surgical outcomes in patients undergoing gastrectomy. Chinese Journal of Cancer Research, 31(5), 797-805. https://doi.org/10.21147/j.issn.1000-9604.2019.05.09

Manara, M., Bona, D., Bonavina, L., & Aiolfi, A. (2024). Impact of pulmonary complications following esophagectomy on long-term survival: Multivariate meta-analysis and restricted mean survival time assessment. Updates in Surgery, 76(3), 757-767. https://doi.org/10.1007/s13304-024-01761-2

Meskers, C. G. M., van der Veen, S., Kim, J., Meskers, C. J. W., Smit, Q. T. S., Verkijk, S., ... van der Leeden, M. (2022). Automated recognition of functioning, activity and participation in COVID-19 from electronic patient records by natural language processing: A proof of concept. Annals of Medicine, 54(1), 235-243. https://doi.org/10.1080/07853890.2021.2025418

Nagabhushan, J. S., Srinath, S., Weir, F., Angerson, W. J., Sugden, B. A., & Morran, C. G. (2007). Comparison of P-POSSUM and O-POSSUM in predicting mortality after oesophagogastric resections. Postgraduate Medical Journal, 83(979), 355-358. https://doi.org/10.1136/pgmj.2006.053223

Peduzzi, P., Concato, J., Kemper, E., Holford, T. R., & Feinstein, A. R. (1996). A simulation study of the number of events per variable in logistic regression analysis. Journal of Clinical Epidemiology, 49(12), 1373-1379. https://doi.org/10.1016/S0895-4356(96)00236-3

Tekkis, P. P., McCulloch, P., Poloniecki, J. D., Prytherch, D. R., Kessaris, N., & Steger, A. C. (2004). Risk-adjusted prediction of operative mortality in oesophagogastric surgery with O-POSSUM. British Journal of Surgery, 91(3), 288-295. https://doi.org/10.1002/bjs.4414

Van Calster, B., McLernon, D. J., van Smeden, M., Wynants, L., & Steyerberg, E. W. (2019). Calibration: The Achilles heel of predictive analytics. BMC Medicine, 17(1), 230. https://doi.org/10.1186/s12916-019-1466-7

van de Beld, J.-J., Crull, D., Mikhal, J., Geerdink, J., Veldhuis, A., Poel, M., & Kouwenhoven, E. A. (2024). Complication prediction after esophagectomy with machine learning. Diagnostics, 14(4), 439. https://doi.org/10.3390/diagnostics14040439 van den Goorbergh, R., van Smeden, M., Timmerman, D., & Van Calster, B. (2022). The harm of class imbalance corrections for risk prediction models: Illustration and simulation using logistic regression. Journal of the American Medical Informatics Association, 29(9), 1525-1534. https://doi.org/10.1093/jamia/ocac093

van Kooten, R. T., Voeten, D. M., Steyerberg, E. W., Hartgrink, H. H., van Berge Henegouwen, M. I., van Hillegersberg, R., Tollenaar, R. A. E. M., & Wouters, M. W. J. M. (2022). Patient-related prognostic factors for anastomotic leakage, major complications, and short-term mortality following esophagectomy for cancer: A systematic review and meta-analyses. Annals of Surgical Oncology, 29(2), 1358-1373. https://doi.org/10.1245/s10434-021-10734-3

Vickers, A. J., & Elkin, E. B. (2006). Decision Curve Analysis: A Novel Method for Evaluating Prediction Models. Medical Decision Making, 26(6), 565-574. https://doi.org/10.1177/0272989X06295361

Voeten, D. M., Busweiler, L. A. D., van der Werf, L. R., Wijnhoven, B. P. L., Verhoeven, R. H. A., van Sandick, J. W., van Hillegersberg, R., & van Berge Henegouwen, M. I. (2021). Outcomes of esophagogastric cancer surgery during eight years of surgical auditing by the Dutch Upper Gastrointestinal Cancer Audit (DUCA). Annals of Surgery, 274(5), 866-873. https://doi.org/10.1097/SLA.0000000000005116

Zhang, G., Liu, X., Hu, Y., Luo, Q., Ruan, L., Xie, H., & Zeng, Y. (2024). Development and comparison of machine-learning models for predicting prolonged postoperative length of stay in lung cancer patients following video-assisted thoracoscopic surgery. Asia-Pacific Journal of Oncology Nursing, 11(6), 100493. https://doi.org/10.1016/j.apjon.2024.100493

Appendix A: Use of Generative AI

This appendix records the use of a generative artificial intelligence assistant in this project, in line with the Utrecht University AI policy for students, under which the permitted level of use for this course is Level 4 (https://www.uu.nl/en/organisation/ai-policy/students/ai-index).

A generative AI assistant was used outside the secure research environment to help develop and debug the analysis code, to structure the analysis, and to draft and refine the wording of written sections. In the course of obtaining this assistance, derived analysis files and aggregate outputs, including a de-identified cohort table, were shared with the assistant, while the modelling of the registry data was carried out within the secure environment. Every numerical result reported in this thesis comes from the author's own runs inside that environment.

All code and text suggestions were reviewed by the author, executed and checked inside the secure environment where they concerned the analysis, and accepted only after verification. The author was mindful that generative AI can produce incorrect or fabricated content and checked the references and numerical claims against primary sources and the author's own outputs.

Appendix B: Code and data availability

The analysis was implemented as a sequence of numbered Python notebooks, covering preprocessing and cohort construction, descriptive statistics, the surgical prediction models, the sensitivity analyses, and the physiotherapy assessment and ICF, together with a script that produces the figures. The code is available at https://github.com/SainiAmandeepSingh/PostoperativeComplicationPrediction. The patient data cannot be shared, as it remains within the secure research environment under the governance of the Dutch Upper Gastrointestinal Cancer Audit and the General Data Protection Regulation; the code is provided so that the analysis can be reproduced on equivalent data. The notebooks in the repository are shared with their executed outputs cleared, so that the full analysis code is visible while no patient-level results leave the secure research environment.

The principal data transformations described in Chapter 3 are reproduced below, lightly edited for readability. The complete pipeline, including the full registry field mappings and the modelling, descriptive, and assessment notebooks, is available in the repository linked above.

B.1 Recoding of registry sentinel values

```
# Registry sentinel codes (9 for binary fields, 99 for scores,
# 999 for measurements) are recoded to missing, per the registry dictionary.
SENT = {'asascore':[9], 'whopreneo':[9], 'neoadj':[9], 'heropn':[9],
        'lengte':[999], 'gewicht':[999], 'gewverl':[999], 'packyears':[999],
        'scorect':[99], 'scorecn':[9], 'conversie':[9], 'anamok':[9],
        'comlong':[9], 'comdiam':[9], 'icdg':[999]}   # full mapping in repo
for c in merged.columns:
    if c.startswith('clav'):
        SENT.setdefault(c, [9])
for col, bad in SENT.items():
    if col in merged.columns and pd.api.types.is_numeric_dtype(merged[col]):
        merged.loc[merged[col].isin(bad), col] = np.nan

# Out-of-range auxiliary measurements are cleared the same way.
AUX = {'packyears':(0,150), 'bloedverlies':(0,10000),
       'duur_operatie':(1,900), 'icdg':(0,365)}
for c, (lo, hi) in AUX.items():
    if c in merged.columns:
        s = pd.to_numeric(merged[c], errors='coerce')
        s = s.where(~s.isin([-95,-96,-97,-98,-99,999,9999,99999]), np.nan)
        merged[c] = s.where((s >= lo) & (s <= hi), np.nan)
```

B.2 Construction of the Charlson Comorbidity Index

```
# Charlson Comorbidity Index from the registry comorbidity fields,
# using the standard condition weights.
def b(col):
    if col not in cohort.columns:
        return pd.Series(0.0, index=cohort.index)
    return (pd.to_numeric(cohort[col], errors='coerce') >= 1).astype(float)

dia = pd.to_numeric(cohort['comdiam'], errors='coerce')
cci = pd.Series(0.0, index=cohort.index)
for c in ['commyo','comhartfaal','comperif','comcva','comdem',
          'comlong','combind','comgiulcus']:
    cci = cci + b(c)                                    # weight 1
cci = cci + np.where(b('compancr') >= 1, 3.0,
                     np.where(b('comlever') >= 1, 1.0, 0.0))
cci = cci + np.where(dia >= 2, 2.0, np.where(dia == 1, 1.0, 0.0))
cci = cci + 2*b('comparalys') + 2*b('comnier')         # weight 2
cci = cci + np.where((b('commalig')+b('leukemie')+b('lymfoom')) >= 1, 2.0, 0.0)
cci = cci + 6*b('comhiv')                               # weight 6
cohort['charlson_cci'] = cci
```

B.3 Extraction of the preoperative physiotherapy risk conclusion

```
# 1. Extract the Conclusie block from each note, bounded by the next 'plan',
#    taking the last occurrence and capping its length.
CONCLUSIE_PATTERN = re.compile(
    r'\bconclusie\b\s*[:\-\.\s]*(.*?)(?=\bplan\b|\Z)',
    flags=re.IGNORECASE | re.DOTALL)
def extract_conclusie(tekst):
    if not isinstance(tekst, str) or not tekst.strip():
        return None
    m = CONCLUSIE_PATTERN.findall(tekst)
    if not m or len(m[-1].strip()) < 5:
        return None
    return m[-1].strip()[:1000]
notes['CONCLUSIE'] = notes['TEKST'].apply(extract_conclusie)

# 2. Classify the risk wording with an ordered keyword scheme (low to high).
RISK_KEYWORDS = [r'laag', r'matig\s+verhoogd', r'matig\s+hoog', r'matig',
                 r'licht\s+verhoogd', r'iets\s+verhoogd',
                 r'sterk\s+verhoogd', r'verhoogd', r'hoog']
RISK_NOUN = r'(?:risico|kans)'
def classify_risk(conclusie):
    if not isinstance(conclusie, str):
        return 'unspecified'
    t = conclusie.lower()
    for kw in RISK_KEYWORDS:
        if re.search(r'\b' + kw + r'\b\s*(?:e?\s+)?' + RISK_NOUN, t):
            return kw
    return 'unspecified'
notes['risk_level'] = notes['CONCLUSIE'].apply(classify_risk)

# 3. Collapse to low/high (matig is low; matig verhoogd and matig hoog are high).
RISK_TO_BINARY = {'laag':'low', 'matig':'low',
                  'matig verhoogd':'high', 'matig hoog':'high',
                  'licht verhoogd':'high', 'iets verhoogd':'high',
                  'verhoogd':'high', 'sterk verhoogd':'high', 'hoog':'high'}

# 4. Carry each conclusion forward in time only (backward as-of merge), so that
#    the conclusion current before surgery is used and no later note can leak in.
merged = pd.merge_asof(notes_dated, events, on='NOTITIE_DATUM_dt',
                       by='MDN', direction='backward')
merged['conc_binary'] = merged.groupby('MDN')['asof_bin'].bfill()
```

B.4 Complete-case fit and cross-validated evaluation

```
# Continuous predictors are standardised inside the pipeline, so the scaler is
# fitted within each training fold only, never on held-out data.
def make_pipe(cols, C=1.0):
    cont = [c for c in CONT if c in cols]
    pre = ColumnTransformer([('s', StandardScaler(), cont)],
                            remainder='passthrough')
    return Pipeline([('t', pre),
        ('lr', LogisticRegression(penalty='l2', C=C, solver='liblinear',
                                  max_iter=5000, random_state=42))])

# The L2 penalty is tuned once on cross validated log loss, then fixed, so it is
# set without reference to the held-out discrimination.
def best_C(X, y):
    cont = [c for c in CONT if c in X.columns]
    pre = ColumnTransformer([('s', StandardScaler(), cont)],
                            remainder='passthrough')
    m = LogisticRegressionCV(Cs=10, cv=5, scoring='neg_log_loss', penalty='l2',
                             solver='liblinear', max_iter=5000, random_state=42)
    Pipeline([('t', pre), ('lr', m)]).fit(X, y)
    return float(m.C_[0])

def calibration_slope(y, p):
    eps = 1e-6; p = np.clip(p, eps, 1 - eps)
    logit = np.log(p / (1 - p)).reshape(-1, 1)
    return float(LogisticRegression(C=1e8, solver='lbfgs', max_iter=3000)
```

```
                 .fit(logit, y).coef_[0, 0])

# Complete-case selection, then stratified 5-fold cross validation with
# held-out predicted probabilities used for every metric.
def evaluate(X, y):
    C = best_C(X, y)
    cv = StratifiedKFold(5, shuffle=True, random_state=42)
    p = cross_val_predict(make_pipe(list(X.columns), C), X, y,
                          cv=cv, method='predict_proba')[:, 1]
    return (roc_auc_score(y, p), brier_score_loss(y, p), calibration_slope(y, p))

cc = df[PREOP + [outcome]].dropna()          # complete case for this model
auc, brier, slope = evaluate(cc[PREOP], cc[outcome].astype(int))
```

Appendix C: Preoperative ICF functioning profile

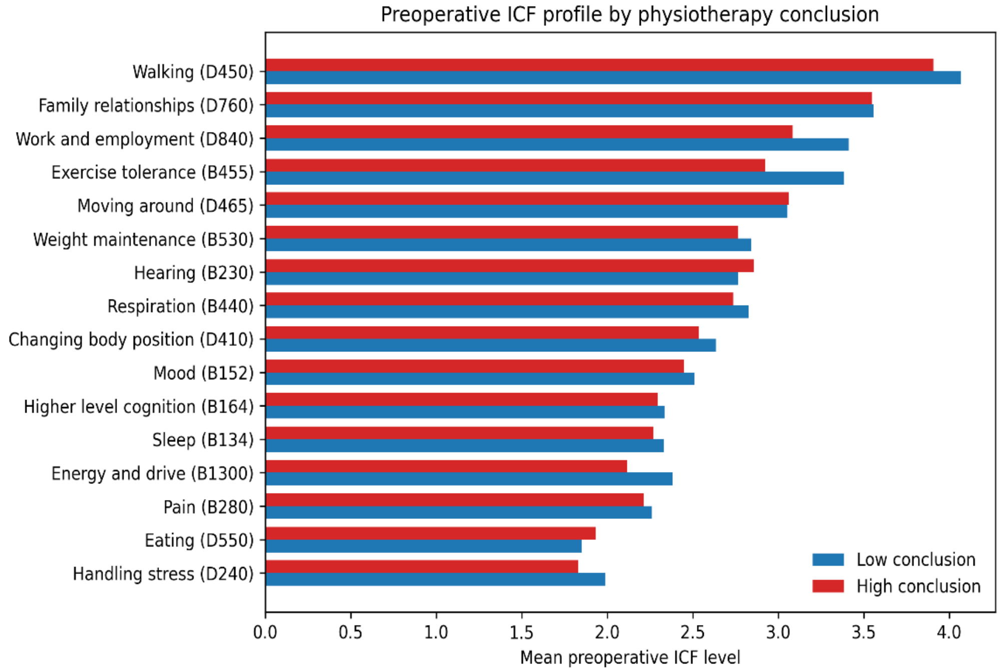

Figure C.1. Mean preoperative ICF functioning level for each category, shown with its International Classification of Functioning code, split by the physiotherapist low and high risk conclusion. Higher values indicate better functioning. Patients judged high risk score lower across most categories, including exercise tolerance.

Appendix D: Preoperative ICF descriptives

Table D.1 summarises the preoperative distribution of each ICF functioning category, and Figure D.1 shows the preoperative course of the six well populated domains. In both, higher values indicate better functioning.

Table D.1. Preoperative distribution of each ICF functioning category, over patients with at least one preoperative value. Higher values indicate better functioning.

ICF category	n with preoperative value	mean	SD	median	IQR
Eating (D550)	483	1.97	1.14	1.8	0.8 to 2.9
Family relationships (D760)	451	3.55	0.43	3.7	3.4 to 3.9
Weight maintenance (B530)	446	2.79	0.62	2.6	2.4 to 3.1
Exercise tolerance (B455)	404	3.16	0.75	3.2	2.6 to 3.9
Walking (D450)	365	3.97	0.39	4.1	3.9 to 4.2
Respiration (B440)	356	2.78	0.81	2.7	2.2 to 3.4
Pain (B280)	336	2.23	0.66	2.0	1.8 to 2.4
Energy and drive (B1300)	301	2.24	0.72	2.0	1.8 to 2.5
Work and employment (D840-D859)	257	3.27	0.67	3.5	3.0 to 3.8
Sleep (B134)	210	2.28	0.68	2.2	1.8 to 2.7
Mood (B152)	196	2.5	0.91	2.2	1.8 to 3.2
Handling stress (D240)	138	1.86	0.45	1.8	1.5 to 2.1
Hearing (B230)	112	2.78	0.73	2.8	2.1 to 3.3
Higher level cognition (B164)	99	2.45	0.83	2.0	2.0 to 2.9
Moving around (D465)	66	3.01	0.6	3.0	2.7 to 3.5
Changing body position (D410)	62	2.6	0.75	2.6	2.1 to 3.0

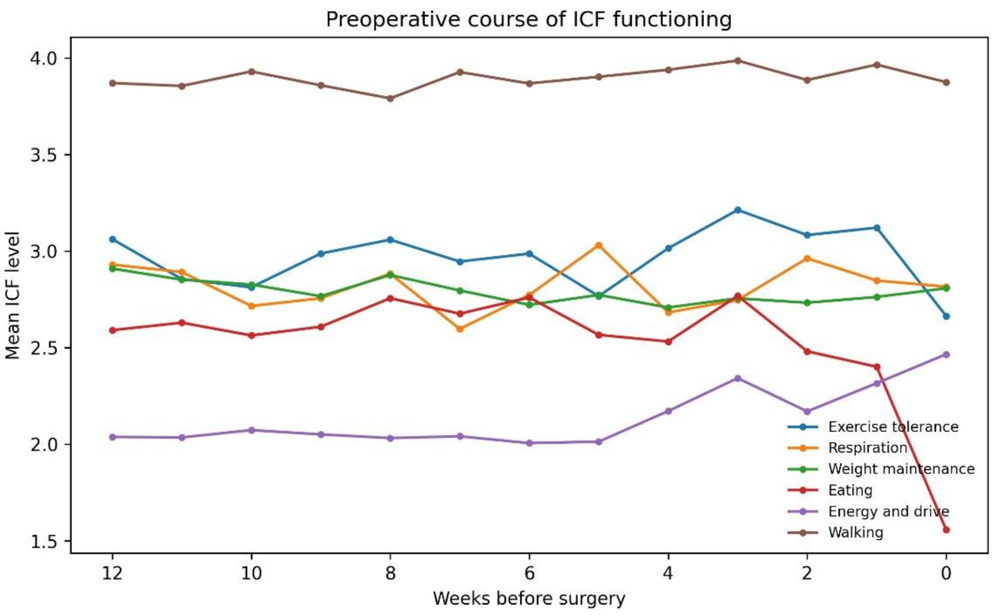

Figure D.1. Mean preoperative ICF level for the six well populated domains, by whole weeks before surgery (0 is the week of surgery). Functioning is broadly stable across the preoperative period; the bins nearest surgery rest on fewer notes and are therefore noisier.